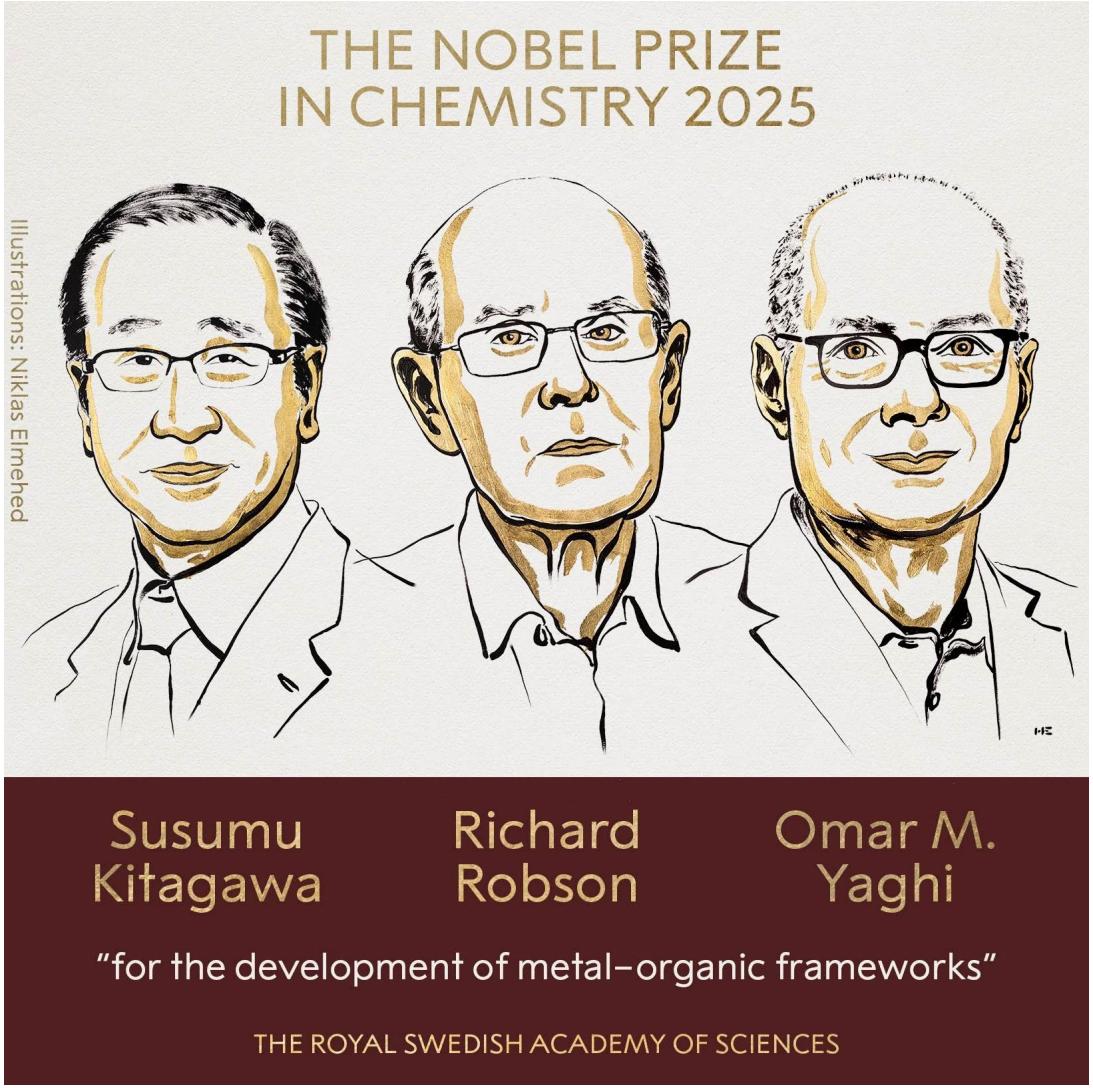
CE 400/500

Special Topics: AI for Chemical Engineers

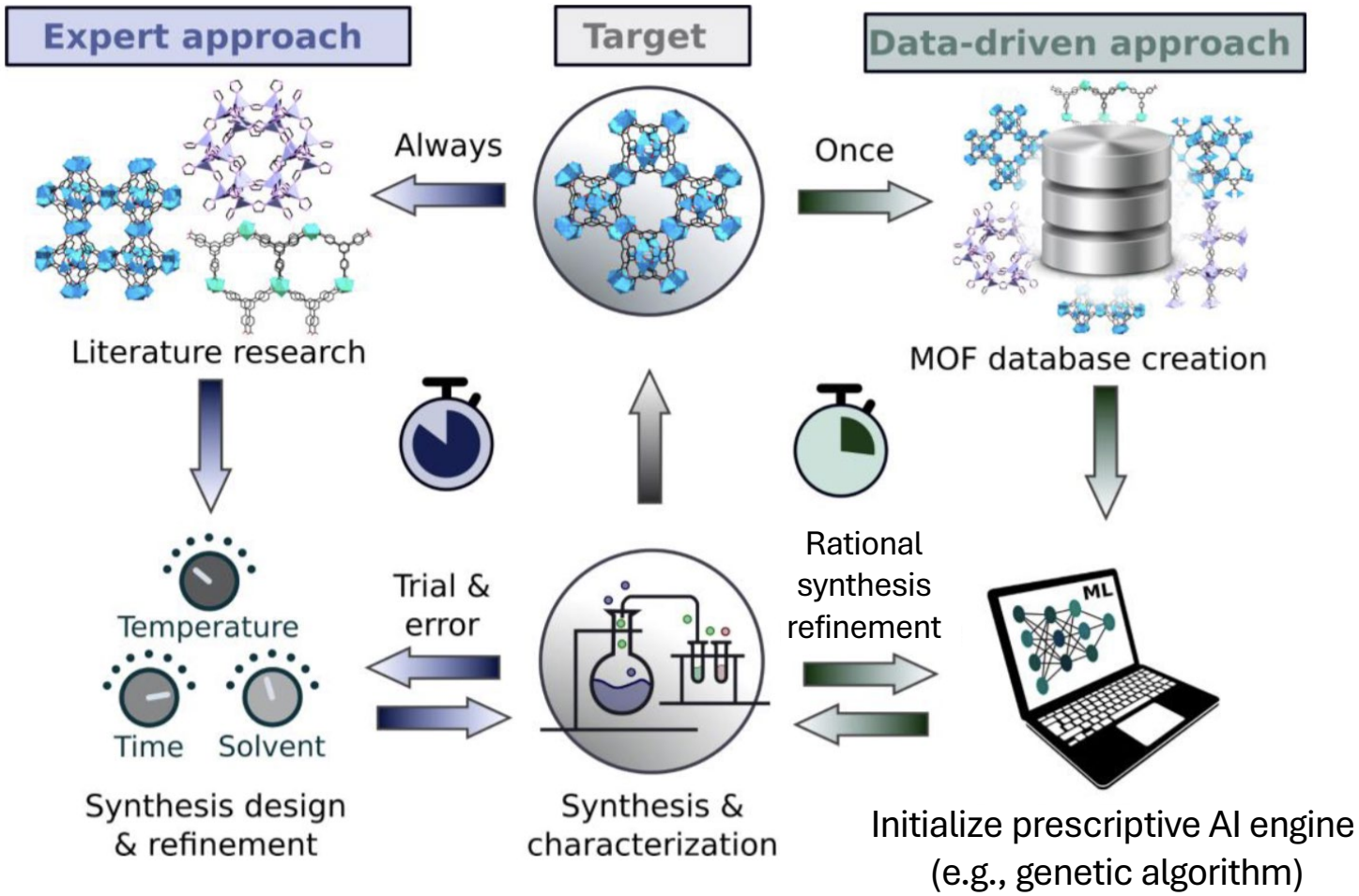
Lecture 17: Hands-on Module: Facilitating the Synthesis of MOFs Using Genetic Algorithm

University at Buffalo
The State University of New York

MOF Won the Nobel Prize in 2025

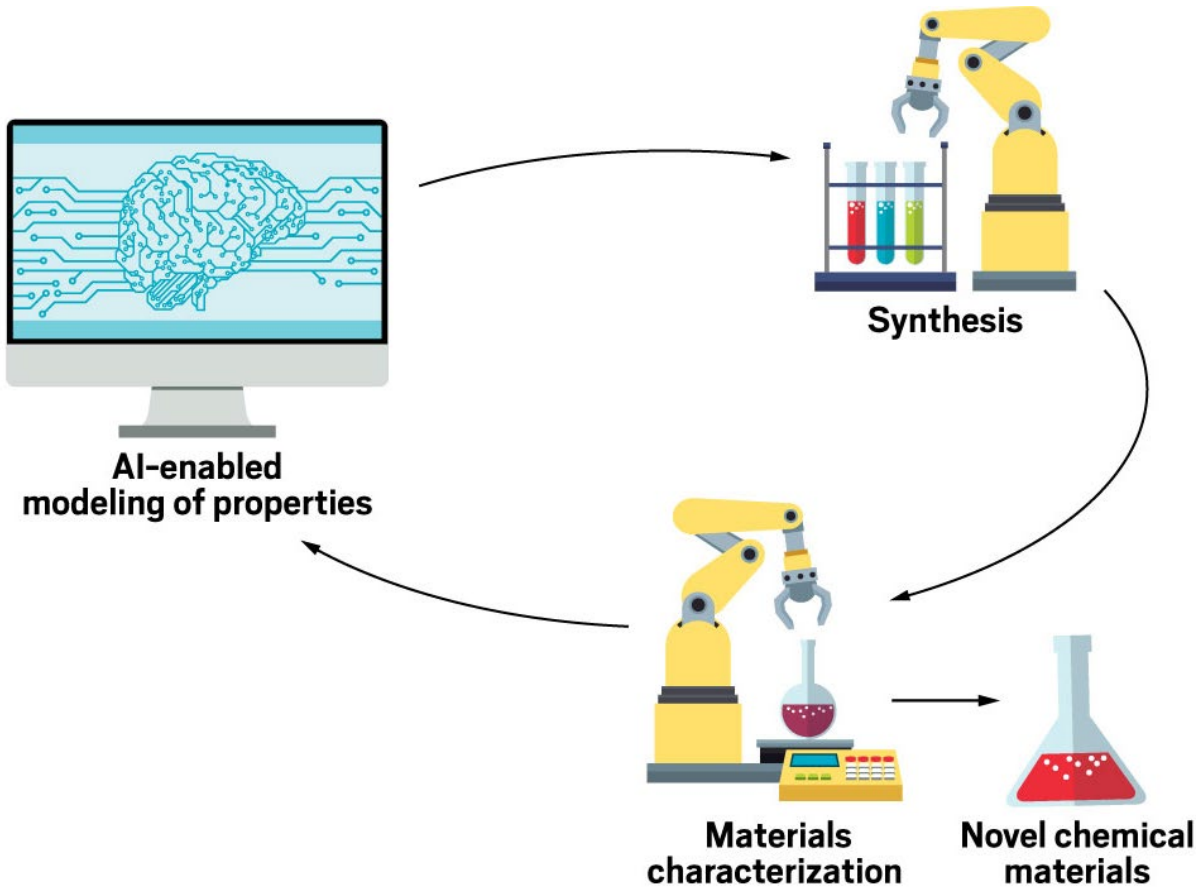


Prescriptive AI can Facilitate Complex MOF Synthesis



Y. Luo, S. Bag, O. Zaremba, A. Cierpka, J. Andreo, S. Wuttke, P. Friederich, and M. Tsotsalas, "MOF Synthesis Prediction Enabled by Automatic Data Mining and Machine Learning," *Angew. Chemie Int. Ed.* 61(19), (2022).

Towards a Fully Autonomous Chemistry Lab



Video demo of robotic chemistry lab

https://cen.acs.org/business/start-ups/CENs-2019-10-Start-Ups-Watch/97/i44

Hands-on: Optimization of Synthesis Conditions for MOFs Using Genetic Algorithm

See Google Colab Notebook